

Paper A-04: Earth Hydrological Systems: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD

Independent Researcher

Founder, Vura Labs

Kampala, Uganda

ORCID: 0009-0001-0556-5516

August 2026

Shared Part IV notation and evidence boundaries are defined in PART_IV_METHODS_AND_CLAIM_BOUNDARY.md. This paper states only its local observables, comparison, and failure conditions.

Abstract

This paper develops a falsifiable CDFD/AFL model of Earth Hydrological Systems. It maps precipitation, snowmelt, groundwater pressure, and gravity to driving flux Φ , channel conveyance, infiltration, storage, and sediment resistance to active constraint C , floodplain storage, wetland buffering, infiltration, and channel adjustment to responsiveness S , and soil moisture, groundwater, channel incision, and sediment legacy to structural memory M_s . The target outcome is discharge, flood extent, avulsion, and recovery time, tested against stream gauges, soil-moisture products, topography, sediment records, and flood maps. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible runtime boundary.

1 Introduction

Water does not merely flow downhill; it actively carves the landscape that channels it. A river network is a living geometry. It evolves, meanders, and periodically destroys its own banks to find a more efficient path to the sea.

In the CDFD ontology, a watershed is not a static pipe system. It is a highly active thermodynamic surface attempting to maximize throughput while minimizing frictional resistance. The shape of a river basin is the physical memory of past fluid flow.

2 The Hydrological Adaptive Surface

We evaluate the dynamics of a river basin using the Adaptive Operating Ratio:

$$\Psi_s = \left(\frac{\Phi}{C} \right) \cdot S \cdot M_s \quad (1)$$

For a specific catchment or river channel:

- **Flux (Φ):** The volumetric flow rate of water (discharge), driven by precipitation and gravity.
- **Constraint (C):** The sheer stress, channel friction, bedrock impermeability, and the topological boundaries of the river valley.
- **Responsiveness (S):** The erodibility of the soil and the capacity of the floodplain to temporarily store excess water.
- **Memory (M_s):** Channel incision. As a river flows over millennia, it cuts deeply into the bedrock. The landscape "remembers" the flow, locking the river into a specific, rigid canyon. This high M_s permanently defines the local constraint geometry.

2.1 Flooding, Avulsion, and Topological Reset

During normal seasonal flow, the river exists at $\Psi_s \approx 1$. The channel geometry (C) perfectly contains and processes the discharge (Φ).

During an extreme precipitation event (e.g., a 100-year storm), the flux Φ spikes exponentially. If the river runs through a deep bedrock canyon (high M_s , locked constraints), the channel cannot flex. The adaptive ratio skyrockets ($\Psi_s \gg 1$). The localized over-flux breaches the physical constraints, leading to a catastrophic flash flood.

In deltaic environments (low M_s , high S), a different phenomenon occurs: **Avulsion**. Over time, sediment deposition slowly clogs the river channel, artificially raising the constraint C and dropping the flow efficiency ($\Psi_s \ll 1$). The river "chokes" on its own memory. To restore thermodynamic equilibrium, the water breaches its levees and carves an entirely new path to the sea. The old constraint structure is abandoned, and a new optimal flow network is established.

3 Measurement Layer

4 Cross-Domain Model Upgrade: Mechanism, Scale, and Tests

4.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Earth Hydrological Systems, the proposed drive is precipitation, snowmelt, groundwater pressure, and gravity. The active limitation is channel conveyance, infiltration, storage, and sediment resistance. Responsiveness is carried by floodplain storage, wetland buffering, infiltration, and channel adjustment, while retained state is represented by soil moisture, groundwater,

channel incision, and sediment legacy. The outcome to explain is discharge, flood extent, avulsion, and recovery time.

Table 1: Operational CDFD/AFL map for Paper A-04.

Term	Paper-specific observable meaning	Measurement question
Φ	precipitation, snowmelt, groundwater pressure, and gravity	What rate, load, gradient, or demand enters the focal system?
C	channel conveyance, infiltration, storage, and sediment resistance	Which measured bottleneck limits transfer, function, or recovery?
S	floodplain storage, wetland buffering, infiltration, and channel adjustment	Which process changes effective capacity or reroutes the load?
M_s	soil moisture, groundwater, channel incision, and sediment legacy	Which retained state makes equal present loads produce different futures?
Y	discharge, flood extent, avulsion, and recovery time	Which outcome is predicted out of sample, with uncertainty?

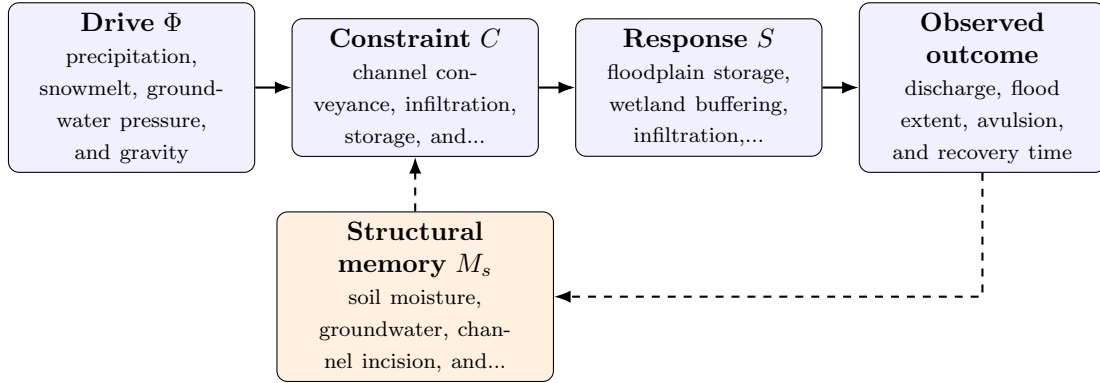


Figure 1: Paper A-04 observable CDFD/AFL architecture for Earth Hydrological Systems. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

4.2 Paper-Specific Causal Hypothesis

For this paper, the hypothesized causal sequence is specific. A change in precipitation, snowmelt, groundwater pressure, and gravity arrives faster than floodplain storage, wetland buffering, infiltration, and channel adjustment can compensate. This raises or redistributes channel conveyance, infiltration, storage, and sediment resistance. If the load persists, the retained state encoded by soil moisture, groundwater, channel incision, and sediment legacy changes the next response, so a later event of equal magnitude can produce a different trajectory in discharge, flood extent, avulsion, and recovery time. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

4.3 Cross-Scale Translation Without Mechanistic Erasure

For the Earth systems family, the cross-domain translation compares coupled transport, finite environmental capacity, buffering, and retained landscape or ecosystem state. It cannot erase conservation laws, geometry, or scale; it must expose where those domain equations create comparable overload and recovery structures. [2, 3, 1] The required scale declaration for this

paper is minutes to millennia; hillslopes to continental basins. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

4.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper A-04. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure stream gauges, soil-moisture products, topography, sediment records, and flood maps and preregister how each record maps to Φ , C , S , M_s , and Y .	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe an extreme rainfall pulse followed by a dry recovery interval while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper’s stated falsifier is: hysteresis and recovery are explained equally well without explicit structural memory.

4.5 Initial One-Site Held-Out Check

The first empirical check is deliberately narrow and reproducible. The local analysis registration and code under `Part_A_Earth_Systems/empirical_hydrology_study/` compare a native lagged-discharge and seasonal ridge baseline with a fixed CDFD/AFL-style extension on daily USGS Potomac River discharge (site 01646500). The chronological 2022–2024 holdout gave an RMSE reduction of only 0.045%, and the 14-day moving-block bootstrap lower bound for mean squared-error reduction was negative. Under the predeclared 1% and positive-lower-bound rule, the extension therefore showed no demonstrated predictive value beyond the native baseline. This result does not test rainfall, measured channel capacity, floodplain storage, or geomorphic memory; it prevents this paper from treating the bookkeeping translation itself as empirical confirmation.

4.6 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from stream gauges, soil-moisture products, topography, sediment records, and flood maps and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of discharge, flood extent, avulsion, and recovery time.

Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the cross-domain translation audit.

5 Conclusion

Hydrology provides a direct test of flow interacting with a surface that the flow itself modifies. CDFD makes river branching comparable to other adaptive transport networks at the level of measurable topology, while the governing hydrological mechanisms remain domain specific.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

6 Reference Layer

References

- [1] Per Bak, Chao Tang, and Kurt Wiesenfeld. Self-organized criticality: An explanation of the $1/f$ noise. *Physical Review Letters*, 59(4):381–384, 1987. DOI: <https://doi.org/10.1103/PhysRevLett.59.381>.
- [2] C. S. Holling. Resilience and stability of ecological systems. *Annual Review of Ecology and Systematics*, 4:1–23, 1973. DOI: <https://doi.org/10.1146/annurev.es.04.110173.000245>.
- [3] A. M. Turing. The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society of London. Series B*, 237(641):37–72, 1952. DOI: <https://doi.org/10.1098/rstb.1952.0012>.